

CCNA / CCNP Network Lab Worksheet

Network Design, Configuration & Validation Workbook

Page 1 — Project Overview

Lab Information

Field	Details
Project Name	
Date Started	
Engineer	
Simulator / Hardware	
IOS / IOS-XE Version	
Objective	

Network Requirements

Purpose of Network

- ☐ Campus LAN
 - ☐ Branch Office
 - ☐ Data Center
 - ☐ WAN Connection
 - ☐ Multi-site Network
 - ☐ Other: ____
-

Required Devices

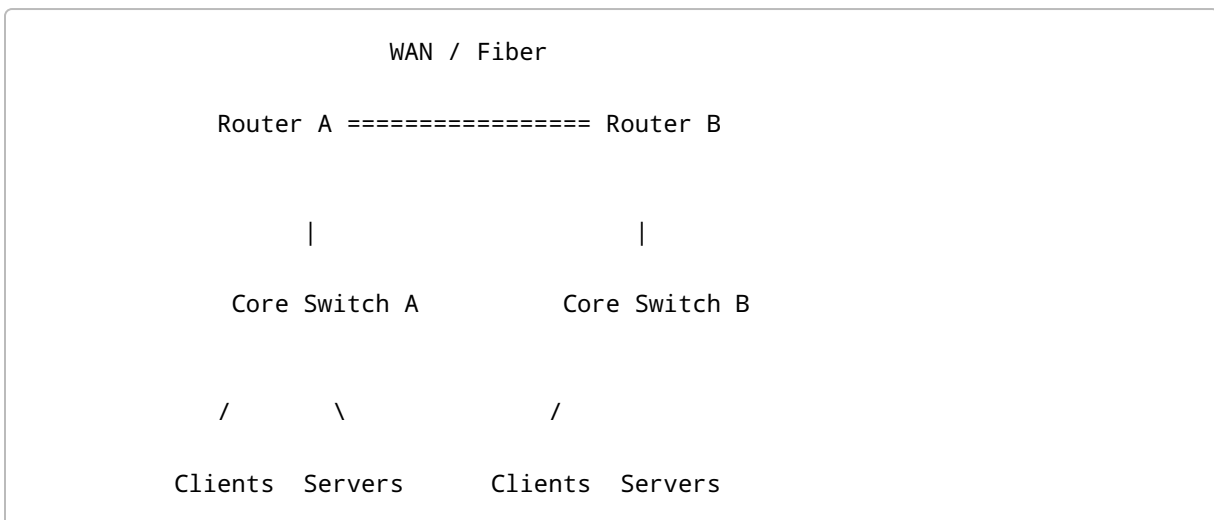
Device Type	Quantity	Model
Routers		
Layer 2 Switches		
Layer 3 Switches		

Device Type	Quantity	Model
Firewalls		
Servers		
End Devices		

Page 2 — Network Topology

Physical Diagram

Draw or attach topology:



Physical Connections

Device A	Interface	Device B	Interface	Cable Type

Page 3 — IP Addressing Plan

Network Summary

Network Purpose	Network Address	Subnet Mask	Notes
WAN / Router Links			
Site A LAN			
Site B LAN			
Management Network			

Router Addressing

Device	Interface	IP Address	Subnet Mask	Description
Router 1				
Router 1				
Router 2				
Router 2				

VLAN Addressing

VLAN ID	VLAN Name	Purpose	Network	Gateway

Page 4 — Device Address Assignment

Infrastructure Devices

Device	Interface	IP Address	Default Gateway	Purpose

End Device Addressing

Device	VLAN	IP Address	Gateway	DHCP / Static

Page 5 — Router Configuration Checklist

Basic Configuration

- ☐ Hostname configured
- ☐ Enable secret configured
- ☐ Console password configured
- ☐ SSH enabled
- ☐ Domain name configured
- ☐ Configuration saved

Interface Configuration

For each interface:

- ☐ Description added
- ☐ IP address configured
- ☐ Interface enabled
- ☐ Link status verified

Commands:

```
show ip interface brief  
  
show interfaces
```

Routing Configuration

Routing Type:

☐ Static Routing

☐ OSPF

☐ EIGRP

☐ BGP

Routing Verification:

```
show ip route  
  
show ip protocols
```

Page 6 — Switch Configuration Checklist

Layer 2 Switching

☐ Hostname configured

☐ VLANs created

☐ Access ports assigned

☐ Trunks configured

☐ Native VLAN configured if required

Verification:

```
show vlan brief  
  
show interfaces trunk
```

Layer 3 Switching

☐ IP routing enabled

```
ip routing
```

☐ SVIs created

☐ Default gateways configured

☐ Routing verified

Verification:

```
show ip interface brief
```

```
show ip route
```

Page 7 — VLAN Design Worksheet

VLAN ID	Name	Purpose	Ports Assigned

VLAN Gateway Information

VLAN	SVI Interface	IP Address

Page 8 — DHCP Configuration Worksheet

DHCP Design

DHCP Provider:

☐ Router

☐ Layer 3 Switch

☐ Dedicated Server

DHCP Scope

Scope Name	Network	Gateway	Range

DHCP Verification

Commands:

```
show ip dhcp binding
```

```
show ip dhcp pool
```

Testing:

☐ Client receives address

☐ Client receives gateway

☐ Client receives DNS

Page 9 — Routing Validation

Routing Table Check

Device	Required Networks	Present?
Router 1		
Router 2		
L3 Switch		

Connectivity Testing

Source	Destination	Result
Device	Gateway	
Router 1	Router 2	
Site A PC	Site B PC	
Client	Server	

Page 10 — Troubleshooting Worksheet

Problem Description

Issue:

Troubleshooting Process

Layer 1 — Physical

☐ Cable/fiber connected

☐ Interface enabled

☐ Link lights active

Commands:

show interfaces

Layer 2 — Switching

☐ Correct VLAN

☐ Correct trunk

- ☐ Correct port mode

Commands:

```
show vlan brief  
  
show interfaces trunk
```

Layer 3 — Routing

- ☐ Correct IP address
- ☐ Correct subnet mask
- ☐ Correct gateway
- ☐ Route exists

Commands:

```
show ip route  
  
ping  
  
tracertoute
```

Page 11 — Documentation Checklist

Before completing the project:

- ☐ Topology diagram saved
- ☐ IP addressing table completed
- ☐ Device configurations exported
- ☐ Screenshots captured
- ☐ Testing results recorded

☐ Troubleshooting documented

☐ Lessons learned written

GitHub Project Structure

```
project-name/  
├── README.md  
├── topology.png  
├── addressing-table.md  
├── configs/  
│   ├── router1.txt  
│   ├── router2.txt  
│   └── switch1.txt  
├── screenshots/  
├── troubleshooting.md  
└── lessons-learned.md
```

Final Network Deployment Checklist

Design

☐ Requirements defined

☐ Topology completed

☐ IP plan created

Configuration

☐ Interfaces configured

☐ VLANs configured

☐ Routing configured

☐ DHCP configured

- ☐ Security configured

Validation

- ☐ Interfaces up/up
- ☐ VLANs verified
- ☐ Routing tables verified
- ☐ End-to-end connectivity tested

Documentation

- ☐ Configuration saved
- ☐ Lab documented
- ☐ Portfolio updated